a

Cell 1 Cell 2 Cell 3 ..... Cell B

Connected

System

Voltage Time

Temperature Time

SOC Time

Operational measurements of Cell 1

Statistics calculation

Voltage std. Time

Maximum voltage Time

Entropy of voltage Time

Statistical time series of battery system

Time series aggregation

b

Voltage Time

Multivariate time-series dataset

Slice partition

Fine-grained time slices

Attribute descriptor assignment

stable decrease stable decrease increase stable

Consolidated semantic descriptor sequences

Descriptor consolidation

stable stable decrease decrease stable stable decrease decrease decrease increase stable stable

Semantics descriptor sequences

Descriptor translation

1~20th time keeps stable, 20~40th time decrease, 40~60th time keeps stable, ……

Translated semantic expressions

Textual expressions

c

You are an expert specializing in time-series signal analysis and maintenance for lithium-ion batteries. Your task is to complete the following “Lithium-ion battery Time-Series Description Text”……

Human instruction

LLM

☐ Voltage of Cell 1:
✓ Trend: The voltage of LIB Cell 1 shows an increasing trend, rising from 3.3413V to 3.3761V over the described period……
✓ Transitions: No significant transition phenomena observed.
✓ Outliers: ……
✓ Variance: ……
✓ ……

Report text of battery operational time series

d

Downstream task application

Task 1: SOC prediction

Cell SOC Time

The predicted SOC at 10 minutes is **9.8%**.....

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Task 2: Operation monitoring

Cell Temperature Time

Temperature **anomaly** is detected……

LLM

Task 3: Charging/discharging management

Cutoff value

Current state

Remaining charging time

Cell SOC Time

Charging can be finished in **5 minutes**……

LLM

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